ON THE SURFACE PRESSURE EQUATION ADVECTION DIFFUSION EQUATION ANALOGY

GITHUB COPILOT

0.1. **Sign-definiteness of diffusivity.** Let

$$\varphi = \sqrt{\frac{if}{\nu\,\mathrm{Ek}}}, \qquad \mathfrak{A} = \frac{\varphi^{-1}\tanh(\varphi H) - H}{\nu\varphi^2 \mathrm{Ek}}, \tag{1}$$

where $H > 0$, $\nu > 0$, $\mathrm{Ek} > 0$, and $f \neq 0$. Introducing

$$\sigma = \mathrm{sgn}(f), \qquad \alpha = \sqrt{\frac{|f|}{2\nu\,\mathrm{Ek}}}, \qquad s = \alpha H > 0, \tag{2}$$

gives

$$\varphi = \alpha(1 + i\sigma), \qquad \nu\varphi^2 \mathrm{Ek} = if. \tag{3}$$

Using

$$\tanh[s(1+i\sigma)] = \frac{\sinh(2s) + i\sigma\sin(2s)}{\cosh(2s) + \cos(2s)}, \tag{4}$$

we obtain

$$\frac{\tanh(\varphi H)}{\varphi} = \frac{\sinh(2s) + \sin(2s) + i\sigma\,[\sin(2s) - \sinh(2s)]}{2\alpha\,[\cosh(2s) + \cos(2s)]}, \tag{5}$$

$$\mathfrak{A}_r \equiv \mathrm{Re}\,\mathfrak{A} = \frac{\sin(2s) - \sinh(2s)}{2\alpha|f|\,[\cosh(2s) + \cos(2s)]}. \tag{6}$$

For $s > 0$,

$$\cosh(2s) + \cos(2s) > 0 \tag{7}$$

and

$$\sinh(2s) - \sin(2s) > 0, \tag{8}$$

the latter following, for example, from

$$\sinh(2s) - \sin(2s) = \int_0^{2s} (\cosh r - \cos r)\; dr > 0. \tag{9}$$

Consequently,

$$\boxed{\mathfrak{A}_r < 0} \qquad (H > 0,\ \nu > 0,\ \mathrm{Ek} > 0,\ f \neq 0). \tag{10}$$

Thus the positive-definite diffusivity is

$$K \equiv -\mathfrak{A}_r = \frac{\sinh(2s) - \sin(2s)}{2\alpha|f|\,[\cosh(2s) + \cos(2s)]} > 0. \tag{11}$$

The pressure-driven transport is therefore strictly down-gradient:

$$\boldsymbol{U}_{p,\mathrm{diss}} = \mathfrak{A}_r \nabla p_s = -K \nabla p_s. \tag{12}$$

Equivalently, the second-order part of the surface-pressure equation is

$$\nabla \cdot (\mathfrak{A}_r \nabla p_s) = -\nabla \cdot (K \nabla p_s), \qquad K > 0. \tag{13}$$

0.2. **Peclet Number.** The surface-pressure equation may be written as

$$-\nabla \cdot (K \nabla p_s) + \boldsymbol{c} \cdot \nabla p_s = S, \tag{14}$$

where

$$K = -\mathfrak{A}_r > 0, \qquad \boldsymbol{c} = (\partial_y \mathfrak{A}_i, -\partial_x \mathfrak{A}_i) = \nabla^{\perp} \mathfrak{A}_i. \tag{15}$$

A natural local Péclet number based on a horizontal length scale L is

$$\boxed{\mathrm{Pe}_L = \frac{L|\boldsymbol{c}|}{K} = \frac{L|\nabla \mathfrak{A}_i|}{K}}. \tag{16}$$

Consequently,

$$\boxed{0 \le \mathrm{Pe}_L < \infty}. \tag{17}$$

In particular, $\mathrm{Pe}_L = 0$ whenever $\mathfrak{A}_i$ is spatially uniform. This occurs, for example, when both H and f are horizontally uniform. There is no universal finite upper bound because $|\nabla \mathfrak{A}_i|$ may be arbitrarily large.

To examine the coefficient dependence, define

$$\alpha = \sqrt{\frac{|f|}{2\nu\,\mathrm{Ek}}}, \qquad s = \alpha H. \tag{18}$$

The positive diffusivity is

$$K = \frac{\sinh(2s) - \sin(2s)}{2\alpha|f|\,[\cosh(2s) + \cos(2s)]}, \tag{19}$$

and the imaginary part of the transport coefficient is

$$\mathfrak{A}_i = \frac{\mathrm{sgn}(f)}{\alpha|f|}\left[s - \frac{\sinh(2s) + \sin(2s)}{2\,[\cosh(2s) + \cos(2s)]}\right]. \tag{20}$$

If $\mathfrak{A}_i$ varies by an amount comparable to $|\mathfrak{A}_i|$ over the horizontal scale L, then

$$L|\nabla \mathfrak{A}_i| \sim |\mathfrak{A}_i|, \tag{21}$$

and hence

$$\mathrm{Pe}_L \sim \frac{|\mathfrak{A}_i|}{K}. \tag{22}$$

The corresponding coefficient ratio is

$$\frac{|\mathfrak{A}_i|}{K} = \frac{|2s\,[\cosh(2s) + \cos(2s)] - \sinh(2s) - \sin(2s)|}{\sinh(2s) - \sin(2s)}. \tag{23}$$

In the shallow or strongly frictional limit,

$$s \ll 1, \tag{24}$$

Taylor expansion gives

$$\frac{|\mathfrak{A}_i|}{K} = \frac{4}{5}s^2 + O(s^6). \tag{25}$$

Thus,

$$\mathrm{Pe}_L \sim \frac{4}{5}s^2 \ll 1. \tag{26}$$

In the deep or weakly frictional limit,

$$s \gg 1, \tag{27}$$

the coefficient ratio satisfies

$$\frac{|\mathfrak{A}_i|}{K} = 2s - 1 + O\left(e^{-2s}\right). \tag{28}$$

Therefore,

$$\mathrm{Pe}_L \sim 2s = H\sqrt{\frac{2|f|}{\nu\,\mathrm{Ek}}}, \qquad s \gg 1, \tag{29}$$

provided that $\mathfrak{A}_i$ has order-one relative variation over L. Hence the effective Péclet number can become very large when the water depth is large compared with the rotational viscous boundary-layer scale:

$$\boxed{\mathrm{Pe}_L \sim \frac{H}{\sqrt{\nu\,\mathrm{Ek}/(2|f|)}} \gg 1}. \tag{30}$$

Large depth alone, however, does not imply large advection. If H and f are horizontally uniform, then

$$\nabla\mathfrak{A}_i = 0, \qquad \boldsymbol{c} = 0, \qquad \mathrm{Pe}_L = 0, \tag{31}$$

regardless of the value of s.

1. Inferred surface pressure velocity and diffusivity fields: redimensionalization

Let

$$\widehat{q} = \sqrt{\frac{i\widehat{f}}{\widehat{\nu}\,\mathrm{Ek}}}, \qquad \widehat{\mathfrak{A}} = \frac{\widehat{q}^{-1}\tanh(\widehat{q}\widehat{H}) - \widehat{H}}{\widehat{\nu}\,\widehat{q}^{\,2}\mathrm{Ek}}. \tag{32}$$

Define

$$\sigma = \mathrm{sgn}(\widehat{f}), \qquad \widehat{\alpha} = \sqrt{\frac{|\widehat{f}|}{2\widehat{\nu}\,\mathrm{Ek}}}, \qquad s = \widehat{\alpha}\widehat{H}, \qquad D(s) = \cosh(2s) + \cos(2s). \tag{33}$$

Then

$$\widehat{q} = \widehat{\alpha}(1 + i\sigma), \qquad \widehat{\nu}\,\widehat{q}^{\,2}\mathrm{Ek} = i\widehat{f}. \tag{34}$$

Writing

$$\widehat{\mathfrak{A}} = \widehat{\mathfrak{A}}_r + i\widehat{\mathfrak{A}}_i, \tag{35}$$

the real and imaginary parts are

$$\widehat{\mathfrak{A}}_r = \frac{\sin(2s) - \sinh(2s)}{2\widehat{\alpha}\,|\widehat{f}|\,D(s)}, \tag{36a}$$

$$\widehat{\mathfrak{A}}_i = \frac{2sD(s) - \sinh(2s) - \sin(2s)}{2\widehat{\alpha}\,\widehat{f}\,D(s)}. \tag{36b}$$

Equivalently,

$$\boxed{\widehat{\mathfrak{A}}_r = \frac{\sin(2s) - \sinh(2s)}{2\widehat{\alpha}\,|\widehat{f}|\,[\cosh(2s) + \cos(2s)]}} \tag{37}$$

and

$$\boxed{\widehat{\mathfrak{A}}_i = \frac{2s\,[\cosh(2s) + \cos(2s)] - \sinh(2s) - \sin(2s)}{2\widehat{\alpha}\,\widehat{f}\,[\cosh(2s) + \cos(2s)]}}. \tag{38}$$

In particular,

$$\widehat{\mathfrak{A}}_r < 0 \qquad \text{for} \qquad \widehat{H} > 0, \quad \widehat{\nu} > 0, \quad \mathrm{Ek} > 0, \quad \widehat{f} \neq 0. \tag{39}$$

The positive nondimensional diffusivity is therefore

$$\widehat{K} \equiv -\widehat{\mathfrak{A}}_r = \frac{\sinh(2s) - \sin(2s)}{2\widehat{\alpha}\,|\widehat{f}|\,[\cosh(2s) + \cos(2s)]} > 0. \tag{40}$$

Use the dimensional scales

$$x_d = L\widehat{x}, \qquad z_d = H_0\widehat{z}, \qquad H_d = H_0\widehat{H}, \tag{41}$$

$$f_d = f_0\widehat{f}, \qquad \nu_d = \nu_0\widehat{\nu}, \qquad \mathrm{Ek} = \frac{\nu_0}{f_0 H_0^2}, \tag{42}$$

and

$$p_{s,d} = \rho_0 f_0 L U_0\,\widehat{p}_s, \qquad \boldsymbol{\mathcal{U}}_d = H_0 U_0\,\widehat{\boldsymbol{\mathcal{U}}}, \tag{43}$$

where $\boldsymbol{\mathcal{U}}_d$ is the dimensional depth-integrated velocity transport.

If

$$\widehat{\boldsymbol{\mathcal{U}}}_p = \widehat{\mathfrak{A}}\,\widehat{\nabla}\widehat{p}_s, \tag{44}$$

then

$$\boldsymbol{\mathcal{U}}_{p,d} = \mathfrak{A}_d\,\nabla_d p_{s,d}, \qquad \mathfrak{A}_d = \frac{H_0}{\rho_0 f_0}\widehat{\mathfrak{A}}. \tag{45}$$

Define

$$q_d = \sqrt{\frac{i f_d}{\nu_d}}, \tag{46}$$

and introduce

$$\sigma_d = \mathrm{sgn}(f_d), \qquad \alpha_d = \sqrt{\frac{|f_d|}{2\nu_d}}, \qquad s_d = \alpha_d H_d. \tag{47}$$

The dimensional Ekman depth is $1/\alpha_d$, so s_d is the fluid depth divided by the Ekman depth.

The parameter s is invariant under the redimensionalization:

$$s = \widehat{\alpha}\widehat{H} = \alpha_d H_d = s_d. \tag{48}$$

The dimensional complex coefficient is

$$\mathfrak{A}_d = \frac{q_d^{-1}\tanh(q_d H_d) - H_d}{\rho_0 \nu_d q_d^2} = \frac{q_d^{-1}\tanh(q_d H_d) - H_d}{i\rho_0 f_d}. \tag{49}$$

Writing

$$\mathfrak{A}_d = \mathfrak{A}_{r,d} + i\mathfrak{A}_{i,d}, \tag{50}$$

its dimensional real and imaginary parts are

$$\mathfrak{A}_{r,d} = \frac{\sin(2s_d) - \sinh(2s_d)}{2\rho_0\alpha_d\,|f_d|\,[\cosh(2s_d) + \cos(2s_d)]}, \tag{51a}$$

$$\mathfrak{A}_{i,d} = \frac{2s_d\,[\cosh(2s_d) + \cos(2s_d)] - \sinh(2s_d) - \sin(2s_d)}{2\rho_0\alpha_d\,f_d\,[\cosh(2s_d) + \cos(2s_d)]}. \tag{51b}$$

Equivalently,

$$\boxed{\mathfrak{A}_{r,d} = \frac{\sin(2s_d) - \sinh(2s_d)}{2\rho_0\alpha_d\,|f_d|\,[\cosh(2s_d) + \cos(2s_d)]} < 0} \tag{52}$$

and

$$\boxed{\mathfrak{A}_{i,d} = \frac{2s_d\,[\cosh(2s_d) + \cos(2s_d)] - \sinh(2s_d) - \sin(2s_d)}{2\rho_0\alpha_d\,f_d\,[\cosh(2s_d) + \cos(2s_d)]}}. \tag{53}$$

The positive dimensional diffusivity coefficient is

$$K_d \equiv -\mathfrak{A}_{r,d} = \frac{\sinh(2s_d) - \sin(2s_d)}{2\rho_0\alpha_d\,|f_d|\,[\cosh(2s_d) + \cos(2s_d)]} > 0. \tag{54}$$

Thus the pressure-driven dimensional transport is

$$\boldsymbol{\mathcal{U}}_{p,d} = -K_d \nabla_d p_{s,d} + \mathfrak{A}_{i,d}\,\boldsymbol{k} \times \nabla_d p_{s,d}. \tag{55}$$

1.1. **Advecting velocity streamfunction decomposition.** The dimensional advecting streamfunction is the imaginary part of $\mathfrak{A}_d$. It separates naturally into inviscid and frictional parts:

$$\mathfrak{A}_{i,d} = \mathfrak{A}_{i,d}^{(0)} + \mathfrak{A}_{i,d}^{(\mathrm{fr})}, \tag{56}$$

where

$$\boxed{\mathfrak{A}_{i,d}^{(0)} = \frac{H_d}{\rho_0 f_d}} \tag{57}$$

is the classical inviscid result, and

$$\boxed{\mathfrak{A}_{i,d}^{(\mathrm{fr})} = -\frac{\sinh(2s_d) + \sin(2s_d)}{2\rho_0\alpha_d f_d\,[\cosh(2s_d) + \cos(2s_d)]}} \tag{58}$$

is the frictional correction, with

$$\alpha_d = \sqrt{\frac{|f_d|}{2\nu_0}}, \qquad s_d = \alpha_d H_d. \tag{59}$$

Thus,

$$\boxed{\mathfrak{A}_{i,d} = \frac{H_d}{\rho_0 f_d} - \frac{\sinh(2s_d) + \sin(2s_d)}{2\rho_0 \alpha_d f_d \left[\cosh(2s_d) + \cos(2s_d)\right]}}. \tag{60}$$

The corresponding advecting velocity is

$$\boldsymbol{c}_d = \nabla_d^\perp \mathfrak{A}_{i,d} = \boldsymbol{c}_d^{(0)} + \boldsymbol{c}_d^{(\mathrm{fr})}, \tag{61}$$

where

$$\boldsymbol{c}_d^{(0)} = \nabla_d^\perp \left(\frac{H_d}{\rho_0 f_d} \right), \tag{62}$$

$$\boldsymbol{c}_d^{(\mathrm{fr})} = -\nabla_d^\perp \left\{ \frac{\sinh(2s_d) + \sin(2s_d)}{2\rho_0 \alpha_d f_d \left[\cosh(2s_d) + \cos(2s_d)\right]} \right\}. \tag{63}$$

Here

$$\nabla_d^\perp \psi \equiv \left(\partial_{y_d}\psi, -\partial_{x_d}\psi\right). \tag{64}$$

Because the frictional correction to the advecting streamfunction is a function of depth H_d and Coriolis parameter f_d, it can only increase or decrease the flow along f_d/H_d contours (not H_d contours). For shallow depths, friction opposes the tendency to flow around f_d/H_d contours: it retards the flow.

1.2. **Physical interpretation.**

1.3. **Advection and diffusion in the surface-pressure equation.**

For uniform dimensional vertical viscosity ν_0, define

$$q_d = \sqrt{\frac{i f_d}{\nu_0}}, \tag{65}$$

and write the dimensional depth-integrated pressure-driven transport as

$$\mathcal{U}_{p,d} = \mathfrak{A}_d \left(\partial_{x_d} + i\partial_{y_d}\right) p_{s,d}, \tag{66}$$

where

$$\mathfrak{A}_d = \frac{1}{i\rho_0 f_d} \left[\frac{\tanh(q_d H_d)}{q_d} - H_d \right]. \tag{67}$$

Here $p_{s,d}$ is dynamic pressure, measured in Pa. If kinematic pressure is used instead, the factors of ρ_0 should be omitted.

Introduce

$$\alpha_d = \sqrt{\frac{|f_d|}{2\nu_0}}, \qquad \sigma_d = \mathrm{sgn}(f_d), \qquad s_d = \alpha_d H_d. \tag{68}$$

Then

$$q_d = \alpha_d(1 + i\sigma_d). \tag{69}$$

Writing

$$\mathfrak{A}_d = \mathfrak{A}_{r,d} + i\mathfrak{A}_{i,d}, \tag{70}$$

the real and imaginary parts are

$$\mathfrak{A}_{r,d} = \frac{\sin(2s_d) - \sinh(2s_d)}{2\rho_0\alpha_d|f_d|\left[\cosh(2s_d) + \cos(2s_d)\right]}, \tag{71a}$$

$$\mathfrak{A}_{i,d} = \frac{2s_d\left[\cosh(2s_d) + \cos(2s_d)\right] - \sinh(2s_d) - \sin(2s_d)}{2\rho_0\alpha_d f_d\left[\cosh(2s_d) + \cos(2s_d)\right]}. \tag{71b}$$

1.3.1. *Advection–diffusion form.* The dimensional surface-pressure equation can be written schematically as

$$-\nabla_d \cdot (K_d \nabla_d p_{s,d}) + \boldsymbol{c}_d \cdot \nabla_d p_{s,d} = S_d, \tag{72}$$

where

$$K_d \equiv -\mathfrak{A}_{r,d} > 0 \tag{73}$$

is the positive pressure-diffusivity coefficient, and

$$\boldsymbol{c}_d = \nabla_d^{\perp}\mathfrak{A}_{i,d} \tag{74}$$

is the advecting velocity. We use the convention

$$\nabla_d^{\perp}\psi \equiv (\partial_{y_d}\psi, -\partial_{x_d}\psi)\,. \tag{75}$$

Because $\boldsymbol{c}_d$ is the perpendicular gradient of a scalar, it is non-divergent:

$$\nabla_d \cdot \boldsymbol{c}_d = \partial_{x_d}\partial_{y_d}\mathfrak{A}_{i,d} - \partial_{y_d}\partial_{x_d}\mathfrak{A}_{i,d} = 0. \tag{76}$$

Moreover,

$$\boldsymbol{c}_d \cdot \nabla_d \mathfrak{A}_{i,d} = 0, \tag{77}$$

so the advecting trajectories follow contours of $\mathfrak{A}_{i,d}$.

1.3.2. *Inviscid and frictional advection.* The advecting streamfunction separates naturally into inviscid and frictional parts:

$$\mathfrak{A}_{i,d} = \mathfrak{A}_{i,d}^{(0)} + \mathfrak{A}_{i,d}^{(\mathrm{fr})}, \tag{78}$$

where

$$\boxed{\mathfrak{A}_{i,d}^{(0)} = \frac{H_d}{\rho_0 f_d}} \tag{79}$$

is the classical inviscid contribution, and

$$\boxed{\mathfrak{A}_{i,d}^{(\mathrm{fr})} = -\frac{\sinh(2s_d) + \sin(2s_d)}{2\rho_0\alpha_d f_d\left[\cosh(2s_d) + \cos(2s_d)\right]}} \tag{80}$$

is the frictional correction.

The corresponding velocity decomposition is

$$\boldsymbol{c}_d = \boldsymbol{c}_d^{(0)} + \boldsymbol{c}_d^{(\mathrm{fr})}, \tag{81}$$

with

$$\boldsymbol{c}_d^{(0)} = \nabla_d^{\perp}\left(\frac{H_d}{\rho_0 f_d}\right), \tag{82a}$$

$$\boldsymbol{c}_d^{(\mathrm{fr})} = \nabla_d^{\perp}\mathfrak{A}_{i,d}^{(\mathrm{fr})}\,. \tag{82b}$$

The inviscid velocity follows contours of H_d/f_d, or, equivalently, contours of f_d/H_d. These are the usual geostrophic contours.

It is useful to define

$$G(s) = \frac{\sinh(2s) + \sin(2s)}{2\left[\cosh(2s) + \cos(2s)\right]}. \tag{83}$$

The magnitude of the frictional streamfunction is then

$$\left|\mathfrak{A}_{i,d}^{(\mathrm{fr})}\right| = \frac{G(s_d)}{\rho_0 \alpha_d |f_d|}. \tag{84}$$

The function G reaches a maximum of approximately

$$G_{\max} \simeq 0.681 \qquad \text{at} \qquad s_d \simeq 0.94, \tag{85}$$

and satisfies

$$G(s_d) \longrightarrow \frac{1}{2} \qquad \text{as} \qquad s_d \longrightarrow \infty. \tag{86}$$

On an f-plane with uniform viscosity, $\mathfrak{A}_{i,d}^{(\mathrm{fr})}$ depends on horizontal position only through H_d. Consequently, its advecting velocity follows isobaths and circulates around topographic features rather than toward or away from them.

In shallow, friction-dominated water,

$$G(s_d) \sim s_d, \qquad s_d \ll 1, \tag{87}$$

and hence

$$\mathfrak{A}_{i,d}^{(\mathrm{fr})} \sim -\frac{H_d}{\rho_0 f_d}. \tag{88}$$

The frictional contribution therefore cancels the inviscid streamfunction to leading order:

$$\mathfrak{A}_{i,d} = \mathfrak{A}_{i,d}^{(0)} + \mathfrak{A}_{i,d}^{(\mathrm{fr})} \longrightarrow 0 \qquad \text{as} \qquad s_d \longrightarrow 0. \tag{89}$$

A shallow, friction-dominated water column therefore cannot sustain the full inviscid depth-integrated geostrophic transport.

For a seamount, the frictional correction modifies the azimuthal advection around the topography. It does not itself produce radial advection toward or away from the summit. In shallow water it opposes and nearly cancels the inviscid advection. In sufficiently deep water the frictional streamfunction becomes nearly independent of depth, so its associated advecting velocity becomes small.

1.3.3. *Sign-definite pressure diffusion.* The positive pressure-diffusivity coefficient is

$$\boxed{K_d = -\mathfrak{A}_{r,d} = \frac{\sinh(2s_d) - \sin(2s_d)}{2\rho_0 \alpha_d |f_d| \left[\cosh(2s_d) + \cos(2s_d)\right]}}. \tag{90}$$

For $s_d > 0$,

$$\cosh(2s_d) + \cos(2s_d) > 0, \tag{91}$$

and

$$\sinh(2s_d) - \sin(2s_d) = \int_0^{2s_d} (\cosh r - \cos r)\, dr > 0. \tag{92}$$

It follows that

$$\boxed{K_d > 0}, \tag{93}$$

or, equivalently,

$$\boxed{\mathfrak{A}_{r,d} < 0}. \tag{94}$$

The pressure-driven diffusive transport is therefore

$$\boxed{\boldsymbol{\mathcal{U}}_{p,d}^{(\mathrm{diff})} = -K_d \nabla_d p_{s,d}}, \tag{95}$$

which is everywhere down the surface-pressure gradient. There is no up-gradient pressure diffusion in this model.

1.3.4. *Dependence of the diffusivity on depth.* Define the universal normalized diffusivity function

$$F(s) = \frac{\sinh(2s) - \sin(2s)}{2\left[\cosh(2s) + \cos(2s)\right]}. \tag{96}$$

Then

$$\rho_0 \alpha_d |f_d| K_d = F(s_d). \tag{97}$$

In shallow or strongly frictional water,

$$F(s_d) = \frac{2}{3} s_d^3 + O(s_d^7), \qquad s_d \ll 1. \tag{98}$$

Therefore,

$$\boxed{K_d \sim \frac{H_d^3}{3\rho_0 \nu_0}}, \qquad s_d \ll 1. \tag{99}$$

The pressure-transport capacity decreases rapidly as the water column becomes shallow. Pressure diffusion is consequently weak over a shallow seamount summit and vanishes as $H_d \to 0$.

At intermediate depth, the normalized diffusivity reaches a modest maximum:

$$F_{\max} \simeq 0.545 \qquad \text{at} \qquad s_d \simeq 1.57. \tag{100}$$

The cross-pressure-contour transport can therefore be most effective on the intermediate-depth flanks of a seamount.

In deep water,

$$F(s_d) \longrightarrow \frac{1}{2}, \qquad s_d \longrightarrow \infty, \tag{101}$$

and

$$\boxed{K_d \longrightarrow \frac{1}{2\rho_0 \alpha_d |f_d|} = \frac{1}{\rho_0 |f_d|} \sqrt{\frac{\nu_0}{2|f_d|}}}. \tag{102}$$

The diffusivity then becomes independent of total depth. Physically, only the bottom Ekman layer participates in the cross-isobar frictional transport, so adding more inviscid water above the bottom boundary layer does not increase the diffusive response.

1.3.5. *Dependence of the diffusivity on the Coriolis parameter.* The dimensional diffusivity can be written explicitly as

$$\boxed{K_d(H_d, f_d) = \frac{\sqrt{2\nu_0}}{\rho_0 |f_d|^{3/2}} F\left(H_d \sqrt{\frac{|f_d|}{2\nu_0}}\right)}. \tag{103}$$

Thus f_d enters both through the prefactor $|f_d|^{-3/2}$ and through the depth parameter

$$s_d = H_d \sqrt{\frac{|f_d|}{2\nu_0}}. \tag{104}$$

The diffusivity is therefore not a function of f_d/H_d alone.

In the shallow or weak-rotation limit,

$$K_d \sim \frac{H_d^3}{3\rho_0 \nu_0}, \tag{105}$$

so its leading-order value is independent of f_d. Rotation is too weak, or the column too shallow, for a distinct bottom Ekman layer to form.

In the deep or strongly rotating limit,

$$K_d \sim \frac{\sqrt{2\nu_0}}{2\rho_0 |f_d|^{3/2}}, \tag{106}$$

so

$$K_d \propto |f_d|^{-3/2}. \tag{107}$$

Pressure diffusion is therefore weaker at large $|f_d|$ and stronger at small $|f_d|$. Strong rotation turns more of the pressure-driven transport along pressure contours and confines the cross-contour response to a thinner bottom Ekman layer.

An identical deep seamount will therefore have stronger pressure-diffusive transport at lower latitude than at higher latitude, provided the planetary-geostrophic approximation remains valid and $f_d \neq 0$.

1.3.6. *Physical interpretation over a seamount.* Diffusion does not have an intrinsically inward or outward direction relative to topography. Its direction is set by the surface-pressure gradient:

$$\boldsymbol{\mathcal{U}}_{p,d}^{(\mathrm{diff})} = -K_d \nabla_d p_{s,d}. \tag{108}$$

Therefore,

- a pressure maximum over a seamount drives outward, down-gradient transport;
- a pressure minimum over a seamount drives inward, down-gradient transport;
- the shallow summit suppresses either transport because $K_d \propto H_d^3$;
- intermediate-depth flanks can support the strongest normalized diffusive transport;
- the deep-water diffusivity becomes independent of depth but varies as $|f_d|^{-3/2}$.

The distinction between advection and diffusion may be summarized as

$$\boxed{\begin{array}{l}\text{Advection follows contours of } \mathfrak{A}_{i,d}(H_d, f_d),\\ \text{diffusion crosses pressure contours with local strength } K_d(H_d, f_d).\end{array}} \tag{109}$$

The classical inviscid advection follows f_d/H_d contours, whereas the frictionally modified advection and the diffusivity are not functions of f_d/H_d alone.